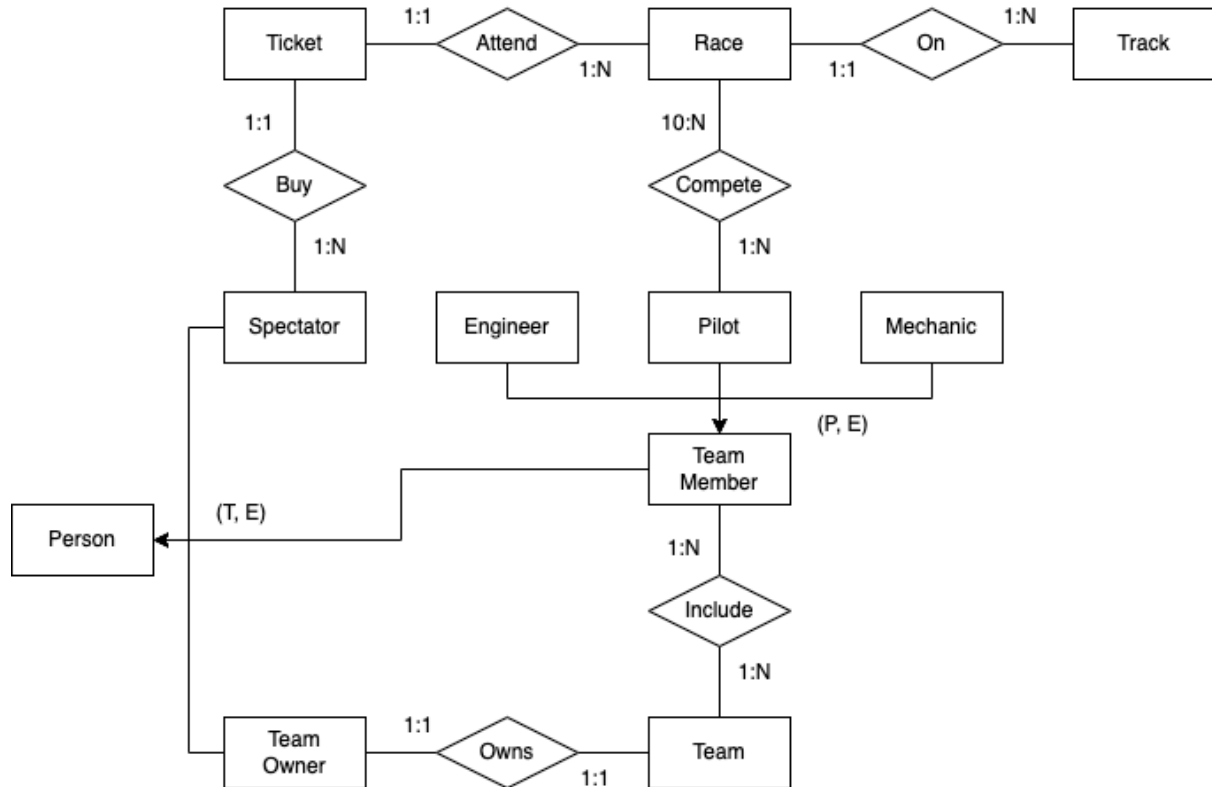


Consider the following ER Diagram, which represents a data structure for tracking Formula 1 races.



The following attributes describe the entities. The primary keys are underlined.

- **Person** - Personal_ID, Name, Surname, Birth_Date, Gender
 - **Spectator** - (...), Favourite_Team
 - **Team Owner** - (...), Total_Assets, Biography (Textual)
 - **Team Member** - (...)
 - **Engineer** - (...), Role_Description (Textual)
 - **Mechanic** - (...), Role_Description (Textual)
 - **Pilot** - (...), Championships_Won, Career (Textual), Career (Textual)
- **Team** - Team_ID, Team_Name, Foundation_Date
- **Ticket** - Ticket_ID, Price, Purchase_Date
- **Race** - Race_ID, Laps_Number, Max_Tickets, Date
- **Track** - Location, Length [Km], Time_Record*

The following attributes describe the relationships. The primary keys are underlined.

- **Include** - Hiring_Date, Resignation_Date
- **Compete** - Winner (Yes/No), Fastest_Lap

N.B. Each Team Member might have worked for different Teams in different periods of time.

Exercise 1 - Neo4j (5 PT)

Consider the Pilot, Race, Track, Ticket, and Spectator entities from the ER model.

Sketch a graph model/example describing the nodes, main attributes, and edges.

Use the proper notation for labels, types, etc. (1 PT)

Write a Cypher query to return the total number of times each pilot achieved a time (*i.e.*, the fastest lap) equal to the track record in races where at least 15 pilots competed before 17/10/2020. (2 PT)

Write a Cypher query to return the total number of races for each track in which at least 1'000 tickets were bought by female spectators, at least 500 of which have Ferrari as their favourite team. (2 PT)

Exercise 2 - Vector DB (1 PT)

Complete the following Zilliz query that considers the pilots whose name starts with “Mic”, and computes how many pilots won each given number of championships.

The results will be in the form: (NumberOfWonChampionships, NumberOfWinningPilots).

Return only the first 5 results.

```
pilots = client.search (  
    collection_name = "pilot_collection",  
    data = [...],  
    search_params = {"metric_type": "IP", "params": {"level": 1}},  
    _____ = _____,  
    _____ = _____,  
    _____ = _____  
)
```

Elasticsearch (3 PT)

Consider the Pilot entity. Provide the complete mapping of the index for the considered entity (i.e., field name, field type, the structure of the mapping, etc.) (1 PT)

Write a query to return male pilots who won at least 10 championships and whose career must include the words “Champion” and “Car”, assigning a higher score to the ones whose career includes the word “Young”. (2 PT)

Last Name (Surname, Cognome) _____ First Name _____ Codice Persona _____

Cassandra (4 PT)

Consider the spectator entity. Write a CQL statement to create its corresponding Cassandra Table using Favourite_Team as a clustering key and Name as an additional partition key (1 PT).

Write a CQL statement to return a list of all the spectators whose name is Bill, whose age is over 65, and whose favourite team is "Ferrari". Besides the query, write any additional CQL statements only if it's strictly necessary. (2 PT)

Write the CQL command to perform a set of statements by loading them from a file located in "D:\Cassandra\Scripts\statement.txt". (1 PT)

MongoDB (5 PT)

Consider the Team Owner, Team, Pilot, Engineer, Mechanic, and Race entities. How many collections would you define within a documental database?

How would you implement the relations between the concepts? Provide a simple documental representation, specifying all attributes and their types or an example. (1 PT)

N.B. Do not include any attributes besides the ones in the ER model.

MONGODB (continued)

NUMBER OF COLLECTIONS: _____ ← **HERE!**

Write a MongoDB query to return the total number of teams with more than 5 pilots and whose owner surname is "Trenti". (2 PT)

Write a MongoDB query to return the total number of teams each pilot competed with, only considering the pilots who ran in at least 100 races in their whole career. (2 PT)